

Using NLTK package to convert audio file to text and text to audio files

AIM: To write a python program to convert audio file to text and text data to audio file.

Description :

NLTK it does not handle audio files directly. We will use speech_recognition and gTTS libraries for audio, and use NLTK to process the text in between.

1. speech_recognition is a Python library (Convert audio file to text)

speech_recognition is a Python library that helps you convert spoken audio into written text using speech recognition engines.

2. gTTS libraries (Convert text file to audio)

gTTS stands for Google Text-to-Speech — it's a Python library and CLI tool that lets you convert text into spoken audio using the Google Text-to-Speech API. It takes in text (string), sends it to Google's Text-to-Speech engine, and returns a spoken audio file in MP3 format.

Install the following packages:

1. !pip install SpeechRecognition pydub
2. !pip install gTTS

Program 1: To convert audio file to text

Audio File : Place it in the same folder as your Python script

```
nltk.download('punkt')
```

```
import speech_recognition as sr
import nltk
```

```
# Initialize recognizer
recognizer = sr.Recognizer()
```

```
# Load audio file
with sr.AudioFile("Sports.wav") as source:
    print("Listening...")
    audio_data = recognizer.record(source)
```

```
# Convert speech to text
try:
    text = recognizer.recognize_google(audio_data)
    print("Recognized Text:", text)
```

```
except sr.UnknownValueError:
    print("Could not understand audio")
```

```
except sr.RequestError:  
    print("API unavailable")
```

OUTPUT:

Recognized Text:

good evening ladies and gentlemen we like to welcome you to play the new videos Broadcast

Tokenized Text:

['good', 'evening', 'ladies', 'and', 'gentlemen', 'we', 'like', 'to', 'welcome', 'you', 'to', 'play', 'the', 'new', 'videos', 'Broadcast']

Saved audio to output.mp3

Program 2 : To convert text to audio file

```
from gtts import gTTS
```

```
text = "Hello, this is generated using Google Text to Speech."
```

```
tts = gTTS(text=text, lang='en')
```

```
# Save audio file
```

```
tts.save("gtts_output.mp3")
```

```
print("Audio saved!")
```

Output:

Audio saved!